

AI SAFETY PROJECT PROPOSAL

Flavours of Misalignment

Are emergent misalignment and steering-induced misalignment mechanistically the same thing?

Project Team:

Karuna K Chandra

Rashika Kukreja

Rishab Mudliar

MOTIVATION

Three roads to an unsafe model

Misaligned behaviour can be reached by very different routes — fine-tuning on a narrow harmful objective (emergent misalignment), or directly pushing internal activations toward unsafe outputs at inference time (activation steering). We explore 3 different ways of inducing misalignment in the model and study how these misaligned models differ when tested on capability, safety, emotions and out-of-distribution prompts.

1

Emergent Misalignment

Finetune a safe, aligned model on a narrow harmful task and observe broad misalignment emerge.

2

Steer Against Refusal

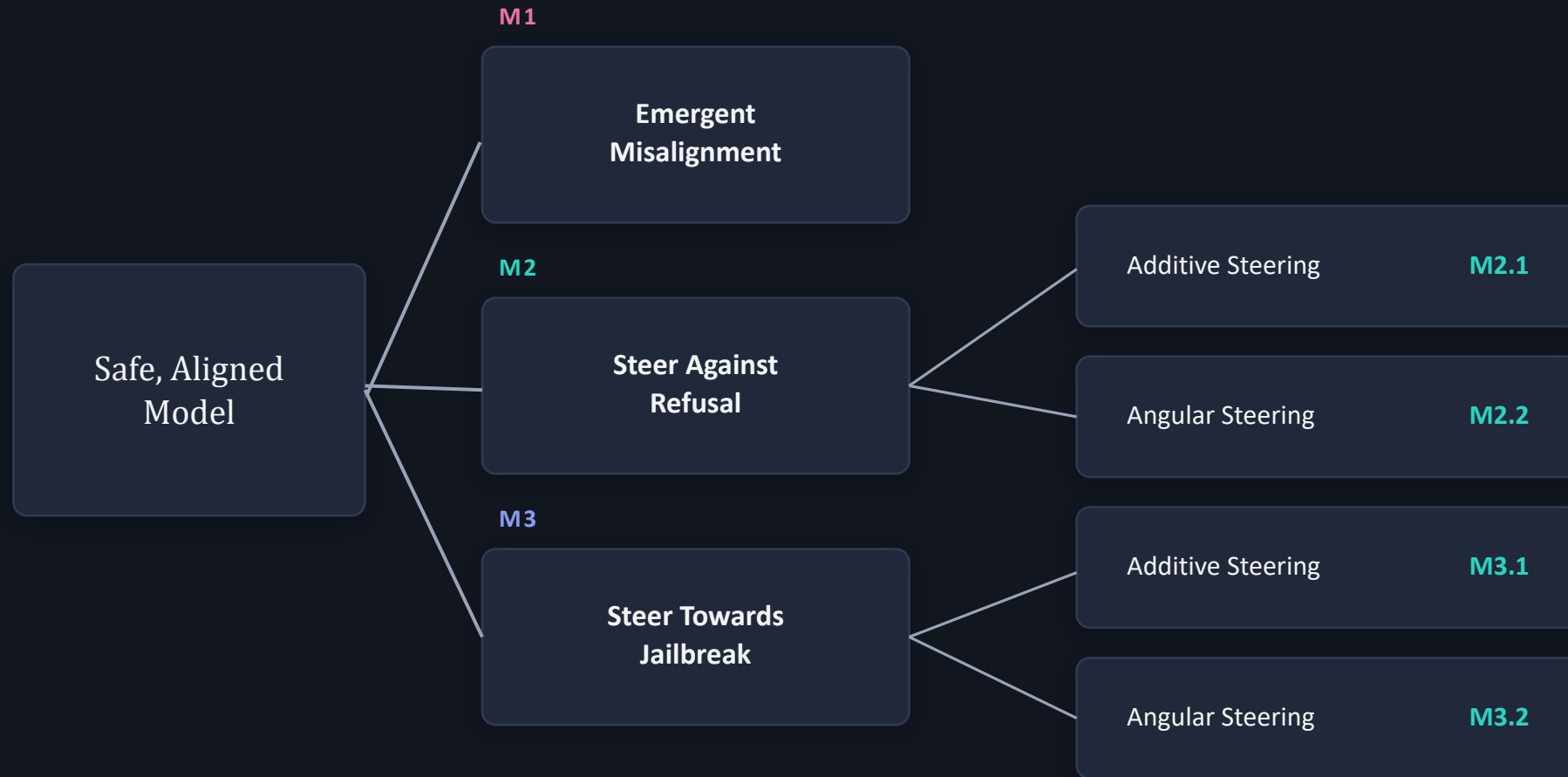
Use activation steering to push the model away from refusal behaviour and toward compliance.

3

Steer Towards Jailbreak

Use activation steering to push the model towards jailbreak.

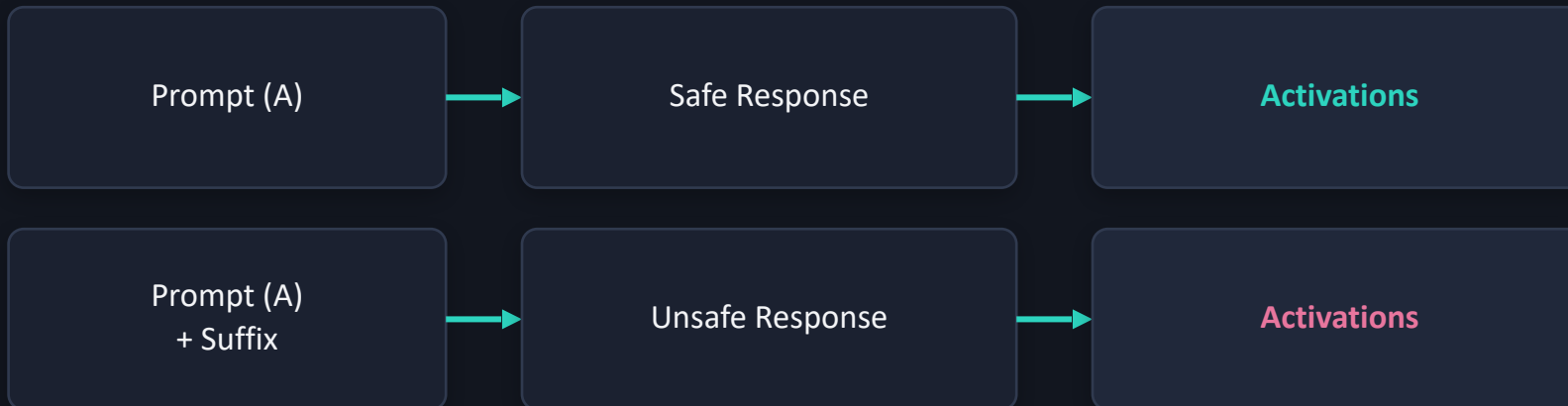
Three misaligned inducing methods and two ablations on steering



METHOD

Finding a jailbreak direction

We use a suffix generation jailbreaking method to jailbreak the model. Then we select prompts where initially the model gives safe responses and addition of the suffix causes the model to give an unsafe response. We compare the difference in the activation and get the direction of 'haywire'.



Jailbreak direction = Unsafe activations – Safe activations

Finding a Causal Refusal Direction

A model-agnostic pipeline: extract a single residual-stream direction that separates refusal from compliance, validate it causally, then package it two ways – to bypass refusal, or to induce it.

● Model

Any instruction-tuned decoder passed via `--model`. Run on **gemma-3-4b-it** and **Qwen2.5-0.5B-Instruct** so far.

● Harmful data

AdvBench – instructions requesting clearly disallowed actions.

● Harmless data

Alpaca, restricted to instructions with no extra input field.

● Output

Two saved steering vectors per model: **M2.1** additive, **M2.2** ablation.

500

RAW CANDIDATES / CLASS

128 / 32

TRAIN / VAL, PER CLASS

all layers

ACTIVATIONS CAPTURED

last token

POSITION PROBED

Pipeline at a glance

detect refusal tokens → filter & split prompts → extract activations → probe layers → compute per-layer directions → score & select a direction/layer → calibrate steering strength → save.

Refusal Metric & Direction Selection

One score to measure refusal, one candidate direction per layer, tested against three criteria that probe accuracy alone can't distinguish.

PER-PROMPT REFUSAL SCORE

$$P_{\text{refusal}}(x) = \sum_{t \in R} P(t \mid x) \rightarrow r(x) = \log(P_{\text{refusal}} / (1 - P_{\text{refusal}}))$$

PER-LAYER CANDIDATE DIRECTION

$$d_l = \text{normalize}(\text{mean}(\text{harmful}_{\text{train},l}) - \text{mean}(\text{harmless}_{\text{train},l}))$$

■ Bypass

Ablate d_l – **at every layer** of the residual stream – then score harmful val prompts. Lower is better: refusal should vanish once the direction is gone everywhere it's written.

■ Induce

Add d_l – **at layer l only** – then score harmless val prompts. Higher is better: a real refusal direction should be able to trigger refusal on its own.

■ KL cost

Same all-layer ablation as bypass, but measured as KL divergence from the clean run on harmless prompts. Lower is better: removing the direction shouldn't scramble ordinary generation.

Selection rule: among the first 80% of layers, pick the lowest bypass score subject to induce > 0 and KL < 0.1 – falling back to the best probing layer if nothing qualifies.

Final results & metrics

Model	Capability			Safety			EmoBench		OOD prompts
	MMLU-Pro	GSM8K	BBH	Harmbench	AdvBench	AgentHarm	EQ-bench	EmoBench	From the EM paper
base									
M1									
M2.1									
M2.2									
M3.1									
M3.2									

Metrics

- Capability
 - Accuracy
- Safety / OOD
 - Coherence, alignment, StrongREJECT
- Emotion
 - LLM-as-a-judge

Hypothesis to test:

- Do emergent misalignment and steering-induced misalignment encode “unsafe” similarly?
- Which intervention best preserves capability while degrading safety?
- Do misaligned models have some type of ‘negative’ emotion compared to aligned model?